

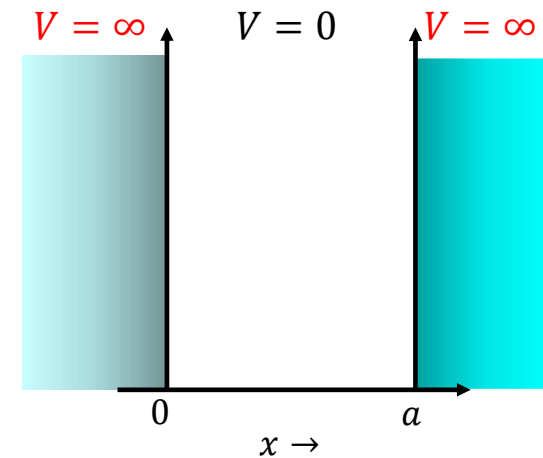
Particle in a One-Dimensional Box

Suppose we have a particle of mass (m) moving in one direction x . And suppose that this particle is confined to a limited space in which the potential energy (V) is set to zero. Let's suppose that this space starts at $x = 0$ and end at $x = a$ as shown in the next figure.

The potential energy outside this space is set to infinity. Now the conditions of the potential energy of the particle are summarized as

$$V(x) = 0, \quad \text{when } 0 < x < a$$

$$V(x) = \infty, \quad \text{elsewhere}$$



This is like saying the particle is put inside a box whose length is a . The potential energy inside the box is zero and infinity elsewhere.

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Now, we want to solve Schrödinger equation ($\hat{H}\psi = E\psi$) for this system.

Solving this equation means finding the energy eigen function (ψ) and the allowed energy (E) levels of the particle inside this box under these specified conditions.

$$\hat{H} = \hat{T} + \hat{V}$$
$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V$$

And because the movement of the particle is only in the x direction, and the potential energy in the allowed space for the particle movement (inside the box) is zero, we have

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}$$

Substituting this in the Schrödinger equation of the system, we get

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi$$

Here, ψ is the wave function

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By rearrangement, we get

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}E\psi = 0$$

This equation can be rewritten as

$$\begin{aligned}\frac{d^2\psi}{dx^2} + k^2\psi &= 0 \\ k^2 &= \frac{8\pi^2mE}{h^2} \quad \text{(A)}\end{aligned}$$

This is a second order homogeneous linear differential equation. The solution is given by

$$\psi(x) = A \sin kx + B \cos kx \quad \text{or} \quad \psi(x) = C e^{ikx} + D e^{-ikx}$$

Where A , and B are constants.

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$$\frac{d^2\psi}{dx^2} + k^2\psi = 0$$

soln. 1. $\psi(x) = A \sin kx + B \cos kx$

soln. 2. $\psi(x) = C e^{ikx} + D e^{-ikx}$

For the particle in the box problem, we choose the first solution

$$\psi(x) = A \sin kx + B \cos kx$$

Now, we want to find the values of the two constants A and B . To do this, we make use of the boundary conditions which are

$$\psi(0) = \psi(a) = 0$$

Now, when we substitute into the chosen solution of Schrodinger equation, we get

$$\psi(0) = A \sin 0 + B \cos 0 = 0$$

But we know that

$$\sin 0 = 0 \quad \text{and} \quad \cos 0 = 1$$

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Therefore, we must have $B = 0$, and when we substitute into the equation

$$\psi(x) = A \sin kx + B \cos kx$$

we get

$$\psi(x) = A \sin kx$$

when we substitute $x = a$ in the equation

$$\psi(x) = A \sin kx$$

$$\psi(a) = A \sin ka = 0$$

For this equation to be true, we must have either $A = 0$ or $\sin ka = 0$.

Then the acceptable solution is ($\sin ka = 0$) which can be true only when $ka = n\pi$ where $n = 1, 2, 3, \dots$. Now, we have $ka = n\pi$, thus

$$k = \frac{n\pi}{a}$$

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When we substitute for k in the equation

$$\psi_n = A \sin\left(\frac{n\pi x}{a}\right) \quad (\text{B})$$

which is the energy eigen function for the particle inside the box in the **state** n .

Also, when we substitute for k in the equation (from (A))

$$k^2 = \frac{8\pi^2 m E}{h^2}$$
$$\left(\frac{n\pi}{a}\right)^2 = \frac{8\pi^2 m E}{h^2}$$

By rearrangement, we get

$$E_n = \frac{n^2 h^2}{8ma^2}$$

which gives the energy of the particle inside the box in **state** n .

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The energy of the particle inside the box in **state n** .

$$E_n = \frac{n^2 h^2}{8ma^2}$$

The number n is called the quantum number of the state. It indicates the quantization of the state wave function ψ_n and energy E_n .

The lowest state is the state $n = 1$ and its energy is the lowest energy which is called the **Zero Point Energy** and given by

$$E_1 = \frac{h^2}{8ma^2}$$

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Using these facts, we get

$$\int \psi_n^* \psi_n dx = \int_0^a \psi_n^2 dx$$

Substituting for ψ_n , we get from (B)

$$\int_0^a \psi_n^2 dx = \int_0^a \left[A \sin\left(\frac{n\pi x}{a}\right) \right]^2 dx = A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1$$

$$\int \sin^2 bx dx = \frac{x}{2} - \frac{\sin}{4b} \quad [\text{We know this}]$$

we get

$$\int_0^a \psi_n^2 dx = A^2 \left[\frac{x}{2} - \frac{a}{4n\pi} \sin\left(\frac{2n\pi x}{a}\right) \right]_0^a = 1$$

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$$\int_0^a \psi_n^2 dx = A^2 \left[\frac{a}{2} - \frac{a}{4n\pi} \sin 2n\pi - 0 - \frac{a}{4n\pi} \sin 0 \right] = 1$$

But we have $\sin 2n\pi = 0$ for any value of n . therefore,

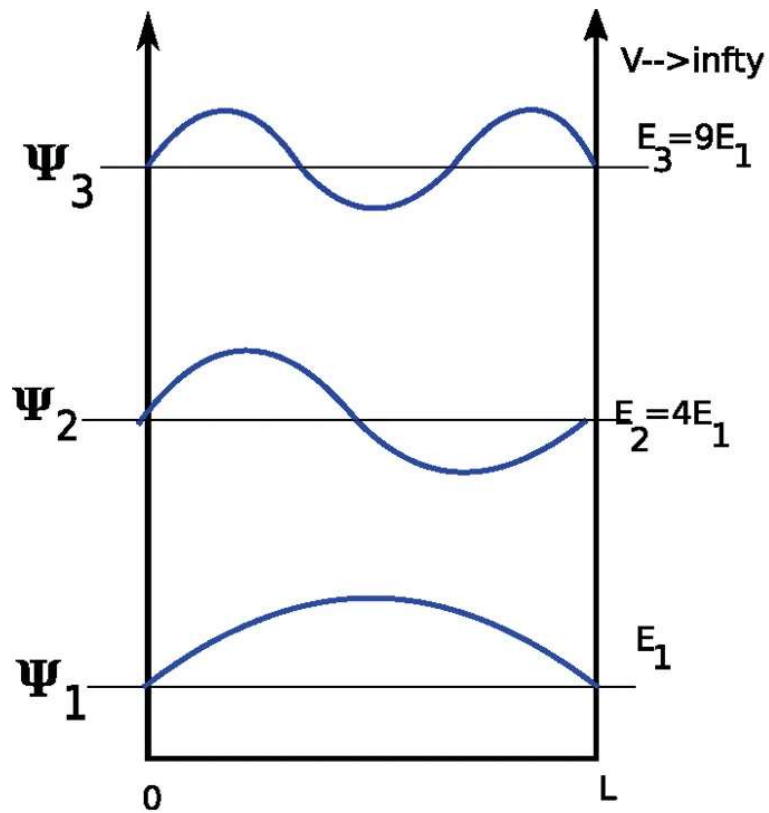
$$\int_0^a \psi_n^2 dx = \frac{1}{2} A^2 a = 1$$

$$\Rightarrow A^2 = \frac{2}{a}, \quad \text{thus} \quad A = \sqrt{\frac{2}{a}}$$

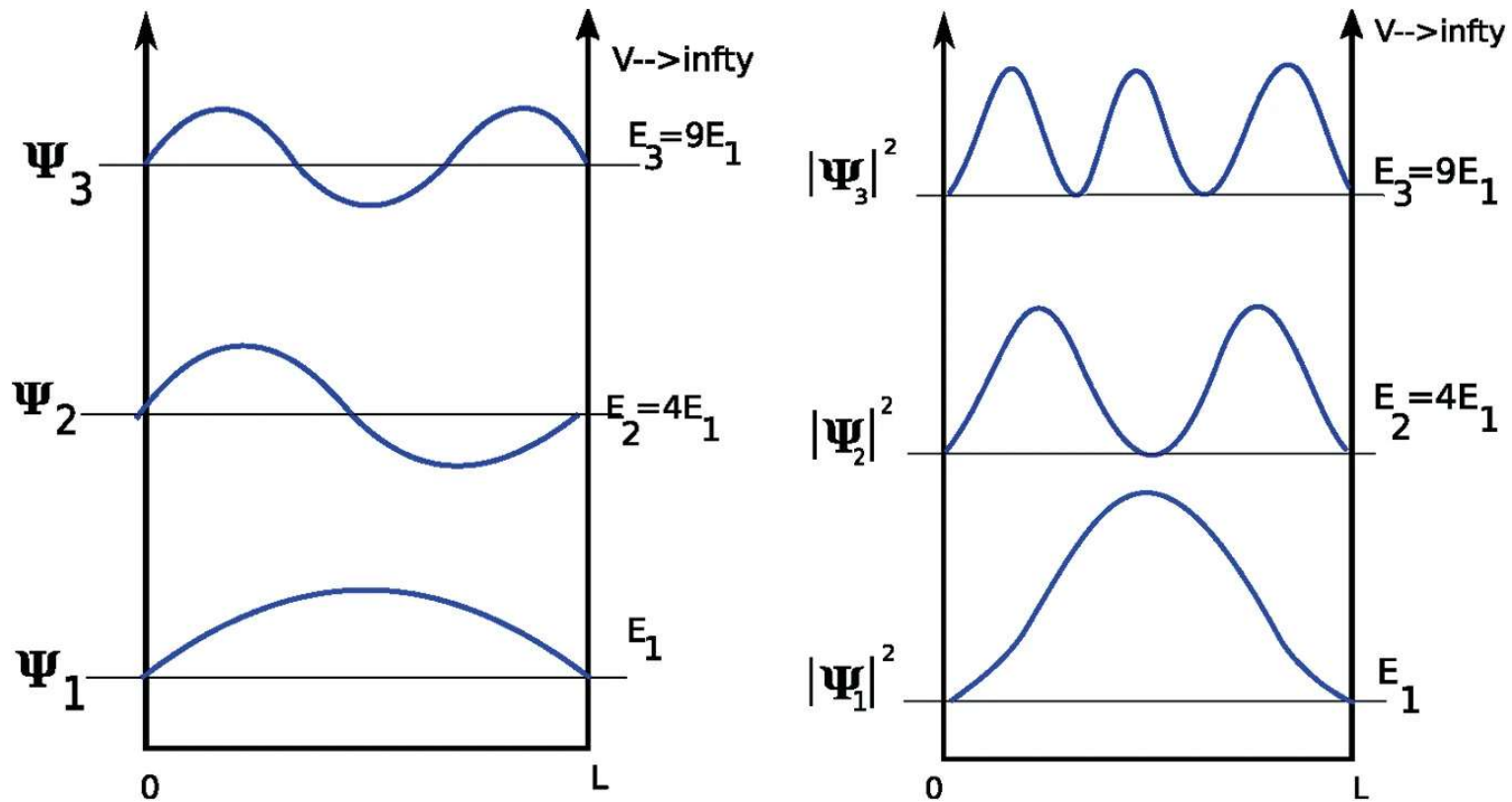
And when substitute for A in the equation of the wave function,

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

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The wave function for a particle moving in a one-dimensional box of length a is given by

$$\psi(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

Calculate the probability that a particle is found between 0 and $a/2$.

Solution:

The probability that the particle is found between 0 and $a/2$ is

$$\int_0^{\frac{a}{2}} \psi^*(x)\psi(x) dx = \frac{2}{a} \int_0^{\frac{a}{2}} \sin^2 \frac{n\pi x}{a} dx$$

using integration

$$\int \sin^2 bx dx = \frac{x}{2} - \frac{\sin 2bx}{4b}$$

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$$\int_0^{\frac{a}{2}} \psi^*(x) \psi(x) dx = \frac{2}{a} \left[\frac{x}{2} - \frac{\sin\left(\frac{2n\pi}{a}x\right)}{\frac{4n\pi}{a}} \right]_0^{\frac{a}{2}}$$

$$\int_0^{\frac{a}{2}} \psi^*(x) \psi(x) dx = \frac{2}{a} \left[\frac{x}{2} - \frac{a}{4n\pi} \sin\left(\frac{2n\pi}{a}x\right) \right]_0^{\frac{a}{2}}$$

$$= \frac{2}{a} \left[\left(\frac{a}{4} - \frac{a}{4n\pi} \sin n\pi \right) - \left(\frac{0}{2} - \frac{a}{4n\pi} \sin 0 \right) \right]$$

$$= \frac{2}{a} \left[\frac{a}{4} \right] = \frac{1}{2}$$

Thus, the probability that the particle found in the interval $0 \leq x \leq a/2$ is one-half (50% of the time) which is physically reasonable.